SentinelMesh

How the defensive investigation platform works

A visual walkthrough of safe flow ingestion, explainable alerts, incident correlation, evidence-grounded reports, and analyst-approved simulations.

Local synthetic data only. No packet capture, external AI provider, shell access, or real network changes.

1. Purpose and system map

SentinelMesh turns prepared network-flow metadata into reviewable investigations. It validates each record, applies a small explainable ruleset, groups findings by source host, saves evidence in SQLite, and builds a report that separates observations from hypotheses.

CLI submits validated flow JSON	Rules create evidence-linked findings	Correlation stores incidents in SQLite
Dashboard reads the loopback API	Investigator writes facts and hypotheses	Analyst approval records a simulation only

Why keep it local and deterministic? The project can run without API keys or third-party services. Each finding points to flow IDs and observed features. A score is only a triage signal; it is not proof of an attack.

2. Run the project

In PowerShell, from the SentinelMesh directory:

```
py -3 -m venv .venv
.\.venv\Scripts\Activate.ps1
.\scripts\run.ps1 analyze
.\scripts\run.ps1 serve
```

Open http://127.0.0.1:8000. The analyze command validates the bundled sample and writes the SQLite database beneath data/generated/. The dashboard can rerun analysis on the bundled sample.

3. Understand a flow and finding

A flow record contains an ID, timestamp, source and destination addresses and ports, protocol, duration, packet count, byte count, and direction. The validator rejects malformed addresses, unsupported protocols, out-of-range counters, duplicate IDs, oversized files, and too many records.

The MVP rules flag outbound connections to commonly sensitive service ports, unusually high byte rates, and repeated multi-port activity. Each finding includes a rule name, severity, score, feature details, and evidence flow IDs.

4. From findings to investigation

Correlation groups findings by source host. The incident keeps linked evidence IDs and the contributing findings, making a path back to observed flow records. The report helper reads only from a short allow-list of local data tools.

Reports separate observed facts from hypotheses. Suggested next steps ask an analyst to verify context. They do not execute shell commands or change network controls.

5. Human approval and audit

The dashboard can record a simulated mark-for-review or host-isolation decision. A named analyst is required. The audit record explicitly says simulated-only and network_changed=false. No firewall, endpoint, or network interface is connected to the project.

6. Folder map

Path	Role
src/sentinelmesh/	Validation, detection, correlation, store, report, API, CLI
data/	Synthetic input; generated SQLite database is ignored
frontend/	Analyst dashboard
docs/	Architecture, safety, contract, walkthrough
scripts/	PowerShell runner and PDF generator
output/pdf/	This illustrated guide

7. Evaluation and limitations

This starter has synthetic examples and heuristic rules, not a labeled benchmark or trained model. To add ML later, split captures by time/session to avoid leakage and report precision, recall, F1, and PR-AUC where appropriate. Measure parsing failures, false positives, and end-to-end latency.

PCAP parsing is intentionally not enabled. The local server has no authentication. Do not expose it beyond localhost or feed it sensitive traffic records. Findings are not proof of malicious activity. Use only benign, owned, or appropriately licensed lab data.